

LESSON 6-8 • SESSION 1 • GUIDED NOTES

Using Properties to Rewrite Expressions

Learning target – I can use the distributive property and like terms to rewrite an expression in simplest form. · 6.EE.A.3

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
Distributive Property <i>propiedad distributiva</i>	Multiply the outside number by ____ term inside the parentheses.	$a(b + c) = ab + ac$
Commutative Property <i>propiedad conmutativa</i>	Changing the ____ when adding or multiplying does not change the answer.	$a + b = b + a$
Associative Property <i>propiedad asociativa</i>	Changing the _____ when adding or multiplying does not change the answer.	$(a + b) + c = a + (b + c)$
like terms <i>términos semejantes</i>	Terms with the same _____ part.	4m and 3m

KEY IDEA*complete each sentence*

- Expressions are _____ when they give the same value for every value of the variable.
- To simplify: distribute, put like terms together, then _____ them.

WORKED EXAMPLE*fill in the blanks as we solve*Simplify $4(m + 2) + 3m$.**Step 1** Distribute: $4(m + 2) = 4m + \underline{\hspace{1cm}}$.**Step 2** Reorder: $4m + 3m + 8$ (Commutative Property).**Step 3** Combine like terms: $4m + 3m = \underline{\hspace{1cm}}$.**Answer** $4(m + 2) + 3m = \underline{\hspace{1cm}}$ **YOU TRY***use the example as a model***1.** Write each expression in simplest form.

a. $2(x + 5) = \underline{\hspace{1cm}}$

b. $3(y + 1) + 2y = \underline{\hspace{1cm}}$

2. Simplify $6(c + 2) + c + 4$. Check your answer with $c = 1$.*Show your work.***Simplest form** _____**SUM IT UP***in your own words*

Two expressions are equivalent when ____, so I can check by ____.

LESSON 6-8 • SESSION 2 • GUIDED NOTES

Expanding and Factoring Expressions

Learning target – I can expand an expression and factor one using the greatest common factor. · 6.EE.A.3

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
expand <i>desarrollar</i>	Multiply to _____ the parentheses.	$6(3x + 4) = 18x + 24$
factor an expression <i>factorizar una expresión</i>	Write it as a _____: a shared factor times a sum.	$18x + 24 = 6(3x + 4)$
term <i>término</i>	One _____ of an expression, separated by + or – signs.	18x and 24 are terms
greatest common factor <i>máximo común divisor</i>	The _____ number that divides both numbers evenly.	GCF of 18 and 24 is 6

KEY IDEA*complete each sentence*

- Factoring _____ expanding, so you can check a factored form by expanding it.
- Factor out the _____ so the terms inside share no other common factor.

WORKED EXAMPLE*fill in the blanks as we solve*Factor $18x + 27$ using the GCF.**Step 1** The GCF of 18 and 27 is _____.**Step 2** Divide each term: $18x \div 9 = \underline{\hspace{1cm}}$ and $27 \div 9 = 3$.**Step 3** Check: $9 \times 2x + 9 \times 3 = 18x + 27$.**Answer** $18x + 27 = \underline{\hspace{2cm}}$ **YOU TRY***use the example as a model***1.** Expand or factor.**a.** Expand $4(2x + 3)$ _____**b.** Factor $15y + 20$ _____**2.** Factor $16n + 24$ using the GCF. Then expand to check.*Show your work.***Factored form** _____**SUM IT UP***in your own words*

To factor an expression, I _____ and I check by _____.
